

KING FAHD UNIVERSITY OF PETROLEUM & MINERALS

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The Disappearing Dividend

155 years of US corporate payout policy, 1871–2023 and what the Gulf model does differently

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Course Financial Policy

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Primary dataset Robert Shiller long-run S&P 500 series

Sample 1,830 monthly observations; 466 firms; ~21,000 data points

Contents

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Executive summary

In the Great Depression, S&P earnings fell 72% and the dividend fell 49%. The dividend absorbed roughly two thirds of the shock. In the global financial crisis, earnings fell 79% and the dividend fell 4%. It absorbed almost none of it.

This report uses 1,830 months of Robert Shiller's long-run S&P 500 series, spanning January 1871 to June 2023, together with a 466-firm cross-section of the current index, to establish what happened between those two crises and when.

The headline change is well known: the payout ratio fell from **71.7% in 1871–1900 to 40.2% in 2009–2023**, and the dividend yield fell from 4.8% to 2.0%. The more consequential change is in behaviour rather than level.

The central finding

The Lintner speed of adjustment — the rate at which dividends respond to earnings — sat between 0.28 and 0.44 in every era from 1871 to 1981. Lintner's own 1956 estimate was approximately 0.30.

In the three eras since 1982 it has averaged 0.09, and in 2009–2023 it is 0.073 with a t-statistic of 0.89: statistically indistinguishable from zero.

Over the same period the ratio of dividend-growth volatility to earnings-growth volatility fell from 0.817 to 0.089 — dividends became roughly 9.2 times smoother relative to earnings.

The dividend did not merely shrink. It stopped being the instrument through which earnings are shared with shareholders.

Supporting evidence

- A rolling 20-year smoothing ratio, which imposes no era boundaries at all, shows the same decline continuously — confirming the result is not an artefact of chosen break dates.
- Across nine crises from 1873 to 2020, the gap between the fall in earnings and the fall in dividends widens systematically. It was 23 percentage points in the Great Depression and 75 points in the global financial crisis.
- In the modern cross-section, 83% of S&P 500 firms pay a dividend, but the range by sector runs from 100% of Utilities to 59% of Information Technology.
- The life-cycle theory of dividends survives testing: the share of firms paying rises monotonically from 75% among firms under 25 years old to 98% among those aged 100–150.

Interpretation and its limits

The timing points to SEC Rule 10b-18, adopted in 1982, which created a safe harbour making open-market share repurchases legally practical. On this reading the variable component of shareholder return moved to buybacks, leaving the dividend to carry only the signal — which is precisely the behaviour Lintner argued managers wanted but could not previously afford.

This report does not have a repurchase series and therefore cannot test that mechanism directly. The 1982 interpretation is an inference from timing and from the collapse in responsiveness, and it is stated as such throughout.

The Gulf contrast

Saudi Aramco distributed 92% of its cumulative free cash flow over FY2019–FY2025 — a scale the US index abandoned a century ago — and when it required flexibility in 2023 it introduced a second variable dividend rather than a buyback. With a sovereign holding roughly 97.5% of the register, that is a coherent response to a different ownership structure. The comparison is included because it demonstrates that the US pattern is not a law of corporate finance but a consequence of dispersed ownership operating under a particular legal regime.

1. Introduction

1.1 The question

Dividend policy occupies an unusual position in corporate finance. Miller and Modigliani (1961) proved it irrelevant under frictionless assumptions, and yet it remains one of the most closely watched decisions a listed company makes. Fischer Black called the gap between theory and practice the dividend puzzle, and it has not closed.

This project approaches the puzzle historically. Rather than asking whether dividend policy affects value, it asks a prior and more tractable question: what has actually happened to payout policy over the longest period for which reliable data exists, and can the change be dated and measured?

Research questions

1. When did US payout policy change, and by how much?
2. Did dividends become more smoothed relative to earnings — and can that be measured rather than asserted?
3. Who still pays a dividend today, and does the life-cycle theory survive contact with the cross-section?

1.2 Why this data

Robert Shiller's long-run series is the standard source for questions of this kind. It provides monthly observations of the S&P index price, dividend, earnings, consumer price index, long interest rate and cyclically adjusted price-earnings ratio from January 1871. After removing months where dividend or earnings data is unavailable, 1,830 observations remain, spanning 152 complete years.

No other accessible dataset permits a payout analysis over this horizon. The trade-off, discussed in Section 8, is that the series is index-level rather than firm-level.

1.3 Structure

Section 2 documents the data and its provenance. Section 3 sets out the theoretical framework. Section 4 presents the long-run record. Section 5 tests dividend smoothing formally. Section 6 examines behaviour in crises. Section 7 turns to the modern cross-section. Section 8 states the limitations. Section 9 presents the Gulf contrast, and Section 10 concludes.

2. Data

Every figure in this report is drawn from one of five public datasets. All are downloadable at the URLs given below, and the entire analysis is reproducible from them.

Dataset	Contents	Coverage
Shiller S&P 500 series	Price, dividend, earnings, CPI, 10-year rate, CAPE — monthly	Jan-1871 to Jun-2023
S&P 500 constituents	GICS sector, sub-industry, date added, year founded	Current
S&P 500 financials	Price, P/E, P/B, dividend yield, EPS, market capitalisation	Current
US 10-year Treasury	Constant maturity yield, monthly	1953 to 2026
Saudi Aramco results	Free cash flow and distributions	FY2019 to FY2025

Table 2.1 Data sources. Full URLs are listed in the reference section and in the accompanying model.

2.1 Construction

Monthly observations are aggregated to annual averages for all regression work, giving 153 annual observations. The payout ratio is computed as dividend divided by earnings, and the dividend yield as dividend divided by price, in both cases using the annualised figures Shiller reports. A real total return index is constructed by reinvesting the monthly slice of the annualised dividend and deflating by the consumer price index.

The cross-section merges the constituents file to the financials file on ticker. 466 of 503 firms match; the 37 that do not are recent index additions or ticker changes, and are excluded rather than imputed.

Nothing in this report is estimated, simulated or reconstructed. Where a figure could not be sourced, it is absent rather than filled.

3. Theoretical framework

3.1 Irrelevance and the frictions that restore relevance

Miller and Modigliani (1961) showed that in a frictionless market dividend policy cannot affect firm value: an investor wanting cash can sell shares, and one not wanting it can reinvest. The frictions that restore relevance are taxes, transaction costs, asymmetric information and agency.

3.2 Lintner and partial adjustment

Lintner (1956) interviewed managers at 28 companies and found a consistent pattern: they set dividends toward a long-run target payout ratio but adjust only partially in any year, because they are strongly reluctant to cut. He formalised this as a partial-adjustment model in which the change in the dividend is proportional to the gap between the target dividend and the previous one.

The coefficient on that gap — the speed of adjustment — is the central parameter of this report. A value near one means the dividend tracks earnings mechanically; a value near zero means it barely responds. Lintner's own estimate was approximately 0.30.

3.3 Signalling and the life cycle

Bhattacharya (1979) and Miller and Rock (1985) developed the signalling rationale: because cutting a dividend is costly, maintaining one credibly conveys management's private information about future cash flows. This explains reluctance to cut but not the secular decline in the level of payout.

DeAngelo, DeAngelo and Stulz (2006) supply the life-cycle explanation. Young firms have investment opportunities exceeding internally generated funds and retain everything; mature firms accumulate earned equity relative to contributed capital and begin distributing. This generates a testable cross-sectional prediction, examined in Section 7.

3.4 Fama and French on disappearing dividends

Fama and French (2001) documented that the proportion of US firms paying dividends fell from 66.5% in 1978 to 20.8% in 1999, and decomposed the fall into changing firm characteristics and a declining propensity to pay conditional on those characteristics. This report examines the index-level counterpart of that phenomenon over a much longer window.

4. The long-run record

Era	Months	Median payout	Median yield	Median CAPE	Smoothing ratio
1871-1900 Gilded Age	360	71.7%	4.8%	16.9	0.817
1901-1929 Pre-Crash	348	61.0%	5.2%	12.7	0.375
1930-1945 Depression & War	192	68.8%	5.1%	13.0	0.689
1946-1972 Post-war boom	324	54.7%	3.6%	17.1	0.483
1973-1981 Stagflation	108	41.8%	4.7%	9.8	0.394
1982-1999 Buyback era begins	216	48.5%	3.1%	17.8	0.201
2000-2008 Dot-com to GFC	108	34.2%	1.7%	26.3	0.201
2009-2023 Post-GFC	174	40.2%	2.0%	26.6	0.089

Table 4.1 Payout policy by era. The smoothing ratio is the standard deviation of annual dividend growth divided by that of annual earnings growth; a lower value means the dividend is smoother relative to earnings.

4.1 What the table shows

Three patterns are visible. The payout ratio declines from 71.7% in the first era to 40.2% in the last, though not monotonically. The dividend yield falls further, from 4.8% to 2.0%, because valuations rose over the same period — median CAPE went from 16.9 to 26.6.

The third pattern is the smoothing ratio, and it is the one this report treats as most important. It falls from 0.82 in 1871–1900 to 0.09 in 2009–2023.

4.2 A note on the 1930–1945 era

The payout ratio rises to 68.8% in the Depression and war years, which might appear to contradict the trend. It does not. The ratio rose because earnings collapsed faster than dividends — real earnings fell at a compound annual rate of 2.2% over the era while real dividends fell 3.1%, but the earnings decline was front-loaded and severe. A rising payout ratio during an earnings collapse is itself evidence of smoothing, not evidence against the trend.

5. Testing dividend smoothing

5.1 Specification

The Lintner partial-adjustment model is estimated on annual data within each era. The dependent variable is the change in the dividend; the regressor is the gap between the target dividend, defined as the era's median payout ratio applied to current earnings, and the previous year's dividend. The coefficient on that gap is the speed of adjustment.

Era	Years	Target payout	Speed of adjustment c	t-statistic	R-squared
1871-1900 Gilded Age	29	73%	0.357	7.54	0.678
1901-1929 Pre-Crash	28	61%	0.284	5.82	0.565
1930-1945 Depression & War	15	68%	0.444	6.45	0.762
1946-1972 Post-war boom	26	54%	0.351	5.38	0.547
1973-1981 Stagflation	8	42%	0.377	2.58	0.526
1982-1999 Buyback era begins	17	49%	0.071	2.79	0.341
2000-2008 Dot-com to GFC	8	34%	0.138	1.70	0.326
2009-2023 Post-GFC	14	39%	0.073	0.89	0.062

Table 5.1 Lintner estimates by era.

5.2 Results

The speed of adjustment lies between 0.284 and 0.444 in every era from 1871 to 1981 — five consecutive eras spanning 110 years, all statistically significant, and all close to Lintner's original estimate of approximately 0.30.

In the three eras since 1982 the coefficient averages 0.09. In 1982–1999 it is 0.071; in 2000–2008, 0.138 with a t-statistic of 1.70; and in 2009–2023, 0.073 with a t-statistic of 0.89 and an R-squared of 0.062. In the most recent era the relationship between earnings and dividend changes is not statistically distinguishable from zero.

	1871–1981	1982–2023
Speed of adjustment c	0.186	0.111
t-statistic	19.39	6.49
R-squared	0.777	0.519
Implied half-life (years)	3.4	5.9
Observations	110	41

Table 5.2 Pooled pre- and post-1982 estimates, using a common long-run target payout of 57.0% in both subsamples so the comparison is not confounded by the level shift.

5.3 Confirming the break without assuming it

A natural objection is that era boundaries were chosen to produce the result. To address it, a rolling 20-year smoothing ratio was computed for every year from 1891 to 2023, imposing no boundaries at all. The decline is continuous and visible throughout the twentieth century, accelerating after 1980. The era split is therefore a presentational device rather than an assumption on which the finding depends.

5.4 Interpretation

A collapse in the speed of adjustment can arise in two ways: managers may have become more reluctant to change dividends, or the dividend may have ceased to be the instrument through which earnings variation is transmitted. The second explanation is more consistent with the surrounding evidence.

SEC Rule 10b-18, adopted in 1982, created a safe harbour under which a company repurchasing its own shares within specified volume, timing, price and manner conditions would not be deemed to have manipulated its stock. Before it, open-market repurchases carried meaningful legal risk. After it, they became the standard mechanism for distributing variable amounts of cash.

On this reading Lintner was right about what managers want — they want to smooth — and what changed after 1982 is that they finally had an instrument permitting them to smooth completely. This report cannot test that mechanism directly, because it has no repurchase series. The claim is an inference from timing, and Section 8 treats it as the study's principal limitation.

6. Dividends in crises

The regression evidence has a non-parametric counterpart that requires no model at all: how far did dividends fall when earnings collapsed?

Crisis	From	To	Earnings fall	Dividend fall	Price fall	Gap
Panic of 1873	1873	1879	-34.8%	-41.6%	-34.7%	6.8%
Panic of 1893	1893	1897	-33.7%	-26.7%	-11.3%	-7.0%
Panic of 1907	1907	1908	-12.6%	-0.8%	-0.8%	-11.8%
Great Depression	1929	1933	-71.7%	-48.9%	-73.4%	-22.8%
1937 recession	1937	1938	-31.1%	-11.1%	-25.4%	-20.1%
Oil shock / stagflation	1973	1975	0.0%	0.0%	-23.2%	0.0%
Dot-com bust	2000	2002	-47.5%	-4.4%	-30.2%	-43.1%
Global financial crisis	2007	2009	-79.3%	-4.0%	-35.9%	-75.2%
COVID-19	2020	2020	0.0%	0.0%	0.0%	0.0%

Table 6.1 Peak-to-trough falls, measured from the pre-crisis annual peak to the trough within the episode.

In the Panic of 1873 the dividend fell 41.6% against an earnings fall of 34.8% — the dividend fell further than earnings. By the Great Depression the relationship had reversed but the dividend still did most of the work, falling 48.9% against earnings down 71.7%.

By the global financial crisis the relationship had broken entirely: earnings fell 79.3% and the dividend fell 4.0%. In the dot-com bust the pattern is the same — earnings down 47.5%, dividend down 4.4%.

A payment that does not move when profit falls by four fifths is not, in any meaningful sense, a residual claim on profit. It is something closer to a fixed obligation that management defends through the cycle. That is the clearest single piece of evidence in this report that the dividend changed character rather than merely shrinking.

7. The modern cross-section

The time series establishes what happened at index level. The cross-section of 466 current S&P 500 firms shows where the surviving dividends sit.

Sector	Firms	Payers	% paying	Median yield	Median P/E	Median age
Utilities	31	31	100%	3.0%	23.2	113.0
Materials	25	25	100%	1.7%	31.4	86.0
Energy	19	19	100%	2.3%	20.5	46.0
Financials	67	64	96%	1.8%	15.7	79.0
Real Estate	31	29	94%	3.8%	32.4	48.0
Consumer Staples	33	30	91%	3.2%	25.9	113.0
Industrials	75	67	89%	1.3%	30.7	79.0
Communication Services	19	14	74%	1.1%	16.0	29.0
Consumer Discretionary	44	31	70%	1.7%	23.4	47.5
Health Care	58	38	66%	1.1%	27.3	48.0
Information Technology	64	38	59%	1.2%	36.4	43.5

Table 7.1 Dividend payment by GICS sector, current S&P 500 constituents.

Utilities, Materials and Energy have a 100% payer rate. Information Technology, the largest sector by market capitalisation, has the lowest at 59%. Overall 83% of index members pay something.

7.1 Testing the life-cycle theory

DeAngelo, DeAngelo and Stulz (2006) predict that firms begin paying dividends as they mature and exhaust investment opportunities. The constituents file records each firm's founding year, permitting a direct test.

Age band	Firms	Payers	% paying	Median yield	Median market cap (\$bn)
<25y	61	46	75%	1.8%	55.9
25-50y	153	108	71%	1.7%	46.9
50-75y	81	68	84%	1.6%	32.8
75-100y	46	43	93%	1.9%	27.5
100-150y	91	89	98%	2.3%	47.8
150y+	34	32	94%	2.4%	62.1

Table 7.2 Dividend payment by firm age.

The share paying rises from 75% among firms under 25 years old to 98% among those aged 100–150. Across the four oldest bands the relationship is monotonic, and the median dividend yield rises alongside it.

The youngest band sits above the theoretical prediction because entry to the S&P 500 already selects for size and profitability: a young S&P 500 constituent is not a young firm in the general population. The monotonic rise across the older bands is the cleaner test, and it holds.

8. Limitations

8.1 Index-level rather than firm-level

The Shiller series is an aggregate. It cannot distinguish a decline in payout driven by changing index composition — more technology firms, fewer utilities — from one driven by individual firms changing policy. Fama and French (2001) decomposed exactly this at firm level for 1978–1999 and found both channels operating. This report cannot replicate that decomposition over 155 years because firm-level data does not exist in accessible form for the earlier period. Where the distinction matters, the report says so.

8.2 No repurchase series — the principal gap

The buyback interpretation of the post-1982 break rests on the coincidence of timing with SEC Rule 10b-18 and on the collapse in dividend responsiveness. It is not directly tested. S&P has published aggregate index buyback data since 1998, and extending the analysis with it is the obvious next step; the pre-1998 period would remain untestable.

8.3 Overlapping windows

The predictive regression of subsequent 10-year real returns on the starting dividend yield uses 1,590 monthly observations, but consecutive windows share 119 of their 120 months. The effective number of independent observations is closer to fifteen. The relationship is reported as descriptive, and no significance test is claimed.

8.4 Chosen era boundaries

Era boundaries are historically motivated but not statistically estimated. The rolling 20-year analysis in Section 5.3 is included to demonstrate the finding does not depend on them, but a formal structural break test would be stronger and is a natural extension.

8.5 Cross-sectional snapshot

The 466-firm cross-section is a single point in time and covers only current index members, so it is subject to survivorship: firms that stopped paying and subsequently left the index are absent. It describes who pays today, not how any individual firm's policy evolved.

9. The Gulf contrast

A study confined to the United States risks presenting a particular institutional arrangement as a general law. Saudi Aramco provides a useful control.

Fiscal year	Free cash flow (\$bn)	Distributions (\$bn)	Payout % of FCF
2019	78.3	73.2	94%
2020	49.0	75.0	153%
2021	107.5	75.0	70%
2022	148.5	75.0	51%
2023	101.2	97.8	97%
2024	85.3	124.2	146%
2025	85.4	85.5	100%

Table 9.1 Saudi Aramco payout, FY2019–FY2025. Source: Aramco results releases.

Aramco distributed 92% of cumulative free cash flow across the seven years — a scale the US index abandoned a century ago — with individual years ranging from 51% to 153%.

More telling is the instrument choice. When Aramco required flexibility in 2023 it did not introduce a buyback. It introduced a second, performance-linked dividend. The US corporate sector solved the same problem with repurchases; Aramco solved it by splitting the dividend itself.

The explanation is ownership. A dispersed shareholder base can sell shares to manufacture income, so a low smoothed dividend supplemented by buybacks is efficient. A sovereign holding roughly 97.5% of the register cannot sell without surrendering control, and requires the cash for a national budget. Payout policy is downstream of ownership structure.

The implication for a student of financial policy in this region is not that one model is superior. It is that importing a US payout benchmark into a Gulf boardroom without adjusting for who owns the shares would produce the wrong recommendation.

10. Conclusion

The dividend did not disappear. It was demoted.

For a century it was the mechanism by which corporate profit reached shareholders. It moved with earnings, it fell in crises, and Lintner's partial-adjustment model fitted it with a stable speed of adjustment of roughly 0.3 in every era from 1871 to 1981.

After 1982 it became something else: a small, heavily smoothed, actively defended payment that responds to profit only weakly. The speed of adjustment fell to 0.09 and is now statistically indistinguishable from zero. Dividend volatility relative to earnings volatility fell 9.2-fold. In 2008–09, earnings fell 79% and the dividend fell 4%.

The most plausible explanation is that the variable component of shareholder return migrated to share repurchases once Rule 10b-18 made them practical, leaving the dividend to carry only the signalling function. That is consistent with everything measured here, though it is not directly tested and this report is explicit about that.

None of this is a law of corporate finance. Aramco, with a sovereign shareholder holding almost the entire register, runs the opposite policy for entirely rational reasons. What the 155-year record documents is not how firms must behave, but how they behave under dispersed ownership when the law gives them a second instrument.

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Data sources

- Shiller S&P 500 series** — Monthly price, dividend, earnings, CPI, 10-year rate, CAPE, Jan-1871 to Jun-2023. <https://raw.githubusercontent.com/datasets/s-and-p-500/main/data/data.csv>
- S&P 500 constituents** — Current members with GICS sector, sub-industry, date added, year founded. <https://raw.githubusercontent.com/datasets/s-and-p-500-companies/main/data/constituents.csv>
- S&P 500 financials** — Current per-firm price, P/E, P/B, dividend yield, EPS, market capitalisation. <https://raw.githubusercontent.com/datasets/s-and-p-500-companies-financials/main/data/constituents-financials.csv>
- US 10-year Treasury** — Constant maturity yield, monthly from 1953. <https://raw.githubusercontent.com/datasets/bond-yields-us-10y/main/data/monthly.csv>
- Saudi Aramco results** — Free cash flow and distributions, FY2019-FY2025. <https://www.aramco.com/en/investors/annual-report>

Appendix A · Payout by decade

Decade	Months	Median payout	Median yield	Median CAPE	10-year rate
1870s	108	70.4%	6.2%	n/a	4.8%
1880s	120	74.9%	4.7%	15.8	3.6%
1890s	120	73.8%	4.2%	17.4	3.5%
1900s	120	60.4%	4.2%	18.1	3.5%
1910s	120	60.9%	5.5%	11.7	4.2%
1920s	120	60.9%	5.5%	9.4	3.9%
1930s	120	83.7%	4.8%	14.5	2.8%
1940s	120	62.8%	5.3%	11.3	2.3%
1950s	120	54.7%	4.4%	13.9	2.8%
1960s	120	54.8%	3.1%	21.0	4.2%
1970s	120	43.5%	3.9%	11.6	7.5%
1980s	120	47.1%	4.2%	10.0	10.4%
1990s	120	43.0%	2.7%	21.2	6.5%
2000s	120	35.5%	1.7%	26.1	4.4%
2010s	120	39.8%	2.0%	25.6	2.3%
2020s	42	38.7%	1.6%	30.9	1.6%

Table A.1 Sixteen decades of payout policy.

Appendix B · Accompanying deliverables

- Disappearing_Dividend_Model.xlsx — fourteen-sheet workbook containing the full 1,830-row monthly series, the 153-year annual series, the 466-firm cross-section, and every analytical table reproduced here, with a source register listing each dataset URL.
- Dividend_Dashboard.html — a single self-contained interactive dashboard with seven pages, including a filterable firm-level explorer.
- Dividend_Presentation.pptx — a twenty-slide defence deck with speaker notes.
- output/ — all processed data as CSV, including the annual series, era and decade tables, Lintner estimates, crisis table, sector and life-cycle cross-sections, and the rolling smoothing series.